

When Closed Forms Exist (and How to Get Them): Green's Functions, Fundamental Solutions, and Classic Constructions

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1 Green's functions in the distributional sense

Definition 1.1 (Fundamental solution). Let L be a linear differential operator on $\mathbb{R}^n$. A *fundamental solution* Φ satisfies $L\Phi = \delta$ in the sense of distributions: for all $\varphi \in C_c^\infty(\mathbb{R}^n)$,

$$\langle \Phi, L^* \varphi \rangle = \varphi(0),$$

where L^* is the formal adjoint of L .

Definition 1.2 (Green's function on a domain). For a bounded domain $\Omega \subset \mathbb{R}^n$ with boundary condition B (e.g. Dirichlet $u|_{\partial\Omega} = 0$ or Neumann $\partial_\nu u|_{\partial\Omega} = 0$), a Green's function $G(\cdot, \xi)$ solves

$$LG(\cdot, \xi) = \delta_\xi \quad \text{in } \Omega, \quad BG(\cdot, \xi) = 0 \quad \text{on } \partial\Omega.$$

Representation formula. If G exists and u solves $Lu = f$ in Ω with boundary data matching B , then

$$u(x) = \int_\Omega G(x, \xi) f(\xi) d\xi + (\text{boundary layer term depending on } B).$$

For homogeneous Dirichlet, the boundary term vanishes.

Jump condition in 1D. If $L = -(py')' + qy$ on (a, b) and $G(\cdot, \xi)$ satisfies $LG(\cdot, \xi) = \delta_\xi$, then G is continuous at $x = \xi$ and its flux has a unit jump:

$$p(\xi) \partial_x G(\xi^+, \xi) - p(\xi) \partial_x G(\xi^-, \xi) = 1.$$

2 1D Sturm–Liouville: constructive recipe

Consider the self-adjoint operator

$$Ly = -(py')' + qy, \quad p > 0, \quad q \text{ real}, \quad x \in (a, b),$$

with self-adjoint boundary conditions (Dirichlet/Neumann/Robin at a, b).

Construction. Let $u_<$ solve $Lu = 0$ satisfying the left boundary condition at $x = a$, and let $u_>$ solve $Lu = 0$ satisfying the right boundary condition at $x = b$. The Wronskian

$$W(x) = p(x)(u_< u_>' - u_<' u_>)$$

is constant ($= W$) on (a, b) . Then

$$G(x, \xi) = \frac{1}{W} \begin{cases} u_<(x) u_>(\xi), & x \leq \xi, \\ u_<(\xi) u_>(x), & x \geq \xi. \end{cases}$$

Example 2.1 (Dirichlet Poisson on $(0, 1)$). $Ly = -y''$, $u(0) = u(1) = 0$. Solutions of $Lu = 0$ with the left/right BC are $u_<(x) = x$ and $u_>(x) = 1 - x$, giving $W = 1$. Thus

$$G(x, \xi) = \min(x, \xi)(1 - \max(x, \xi)) = \begin{cases} x(1 - \xi), & x \leq \xi, \\ \xi(1 - x), & x \geq \xi. \end{cases}$$

Example 2.2 (Dirichlet with $Ly = -y''$ on $(0, 1)$ but hyperbolic basis). Setting $u_< = \sinh x$, $u_> = \sinh(1 - x)$ also yields $W = \sinh 1$, hence

$$G(x, \xi) = \frac{1}{\sinh 1} \begin{cases} \sinh x \sinh(1 - \xi), & x \leq \xi, \\ \sinh \xi \sinh(1 - x), & x \geq \xi. \end{cases}$$

Remark 2.3 (Robin and Neumann). For Robin $a_0 u + a_1 p u' = 0$ at an endpoint, pick $u_<, u_>$ to satisfy those mixed conditions. The same Wronskian formula applies; only W changes.

3 Free-space fundamental solutions in $\mathbb{R}^n$

Let $r = |x|$ and $H_0^{(1)}$ be the Hankel function.

Laplace $-\Delta \Phi = \delta$:

$$\Phi_n(x) = \begin{cases} \frac{1}{(n-2)\omega_n} r^{2-n}, & n \geq 3, \\ -\frac{1}{2\pi} \log r, & n = 2, \end{cases} \quad \omega_n = \text{surface area of } S^{n-1} = \frac{2\pi^{n/2}}{\Gamma(n/2)}.$$

Helmholtz $(\Delta + k^2)\Phi = \delta$ (outgoing):

$$\Phi_3(x) = \frac{e^{ikr}}{4\pi r}, \quad \Phi_2(x) = \frac{i}{4} H_0^{(1)}(kr).$$

Remark 3.1. These are distributional identities; away from $r = 0$ the kernels are classical solutions. The singularity encodes the point source.

4 Method of images

For domains with simple symmetries, combine free-space kernels with *images* that enforce boundary conditions.

Half-space Dirichlet for Laplace (3D). For $\Omega = \{x_3 > 0\}$, Dirichlet data $u|_{x_3=0} = 0$:

$$G(x, \xi) = \frac{1}{4\pi} \left(\frac{1}{|x - \xi|} - \frac{1}{|x - \xi^*|} \right), \quad \xi^* = (\xi_1, \xi_2, -\xi_3).$$

Neumann uses a plus sign.

Sphere (Kelvin transform). For a ball, the image point is at the inversion point and a scaling is needed; the same idea applies with a modified coefficient.

5 Conformal mapping in 2D

If $\phi : \Omega \rightarrow \Omega'$ is conformal and $G_{\Omega'}$ is the Dirichlet Green's function on Ω' , then

$$G_{\Omega}(z, \xi) = G_{\Omega'}(\phi(z), \phi(\xi))$$

because the Laplacian is conformally invariant up to a Jacobian that cancels in the Green identity for Dirichlet problems.

Unit disk Dirichlet kernel. For $\mathbb{D} = \{z : |z| < 1\}$,

$$G_{\mathbb{D}}(z, \xi) = -\frac{1}{2\pi} \log \left| \frac{z - \xi}{1 - \bar{\xi}z} \right|.$$

Example 5.1 (Upper half-plane). Map $\mathbb{H} = \{\text{Im } z > 0\}$ to $\mathbb{D}$ by the Cayley transform $\phi(z) = \frac{z - i}{z + i}$. Then

$$G_{\mathbb{H}}(z, \xi) = G_{\mathbb{D}}(\phi(z), \phi(\xi)) = -\frac{1}{2\pi} \log \left| \frac{z - \xi}{z - \bar{\xi}} \right|.$$

This equals the method-of-images kernel for the line boundary.

6 Existence, uniqueness, and the Green identity

Theorem 6.1 (Uniqueness for Dirichlet Poisson). *Let Ω be bounded with C^1 boundary. For $Lu = -\Delta u$ with Dirichlet data, the Green's function is unique, symmetric $G(x, \xi) = G(\xi, x)$, and*

$$u(x) = \int_{\Omega} G(x, \xi) f(\xi) d\xi$$

is the unique weak solution in $H_0^1(\Omega)$ for $f \in H^{-1}(\Omega)$.

Sketch. Lax–Milgram yields existence/uniqueness in H_0^1 . Symmetry follows from self-adjointness and the reciprocity identity $\int G(x, \xi) f(\xi) d\xi = \int G(\xi, x) f(\xi) d\xi$ for all f . $\square$

Weak formulation. For $f \in H^{-1}(\Omega)$, $u \in H_0^1(\Omega)$ solves $-\Delta u = f$ iff

$$\int_{\Omega} \nabla u \cdot \nabla \varphi dx = \langle f, \varphi \rangle \quad \forall \varphi \in H_0^1(\Omega).$$

With G , testing against $\varphi(\cdot) = G(\cdot, \xi)$ reproduces the representation.

7 Eigenfunction expansions on bounded domains

Let $\Omega \subset \mathbb{R}^n$ be bounded with sufficiently regular boundary and let

$$Lu := -\text{div}(A \nabla u) + c u$$

be a uniformly elliptic, real, self-adjoint operator on $H_0^1(\Omega)$ (Dirichlet) or on $H^1(\Omega)$ with Neumann/Robin boundary conditions. Assume $A(x)$ is symmetric, bounded, and uniformly positive definite, $c \in L^\infty(\Omega)$ with $c \geq 0$.

Theorem 7.1 (Spectral theorem (compact resolvent)). *There exists an orthonormal basis $\{\phi_k\}_{k=1}^\infty$ of $L^2(\Omega)$ consisting of eigenfunctions of L ,*

$$L\phi_k = \lambda_k \phi_k, \quad 0 < \lambda_1 \leq \lambda_2 \leq \dots, \quad \lambda_k \rightarrow \infty,$$

for Dirichlet (and $\lambda_1 = 0$ may occur for Neumann/Robin, with ϕ_1 constant when $c \equiv 0$). The eigenfunctions are in $H^2(\Omega)$ (and smooth if $A, c, \partial\Omega$ are smooth).

Green series. For Dirichlet (or after removing the nullspace for Neumann/Robin), the Green's function admits the *eigenfunction expansion*

$$G(x, \xi) = \sum_{k=1}^{\infty} \frac{\phi_k(x) \phi_k(\xi)}{\lambda_k}, \quad (1)$$

converging in $L^2(\Omega \times \Omega)$ and, away from the diagonal $x = \xi$, locally uniformly. Termwise application of L gives $LG(\cdot, \xi) = \delta_\xi$ in the distributional sense.

Remark 7.2 (Neumann/Robin zero mode). If 0 is an eigenvalue (e.g. Neumann with $c \equiv 0$), use the *reduced Green's function* by projecting off the nullspace:

$$G(x, \xi) = \sum_{\lambda_k > 0} \frac{\phi_k(x) \phi_k(\xi)}{\lambda_k}.$$

The representation of solutions then holds for data orthogonal to the nullspace (e.g. zero-mean f for Neumann Laplacian).

Example 7.3 (Rectangle, Dirichlet Laplacian). $\Omega = (0, a) \times (0, b)$, $L = -\Delta$, $u|_{\partial\Omega} = 0$. The eigenpairs are

$$\phi_{mn}(x, y) = \frac{2}{\sqrt{ab}} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}, \quad \lambda_{mn} = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2, \quad m, n \in \mathbb{N}.$$

Hence

$$G(x, \xi) = \sum_{m, n \geq 1} \frac{\phi_{mn}(x) \phi_{mn}(\xi)}{\lambda_{mn}}.$$

This is a sine series; for strips and intervals the sum collapses to closed forms with $\sinh / \cosh$.

Example 7.4 (Disk, Dirichlet Laplacian). In polar coordinates, the eigenfunctions are $J_{|m|}(j_{|m|,k}r/R)e^{im\theta}$, with $\lambda_{m,k} = j_{|m|,k}^2/R^2$ (zeros of Bessel $J_{|m|}$). Then

$$G(r, \theta; \rho, \vartheta) = \sum_{m \in \mathbb{Z}} \sum_{k \geq 1} \frac{J_{|m|}(j_{|m|,k}r/R) J_{|m|}(j_{|m|,k}\rho/R)}{\lambda_{m,k} \pi R^2 [J_{|m|+1}(j_{|m|,k})]^2} e^{im(\theta - \vartheta)}.$$

Remark 7.5 (Convergence and regularity). The series (1) converges absolutely on compact subsets of $\{(x, \xi) : x \neq \xi\}$ and reproduces the logarithmic (2D) or Newtonian ($n \geq 3$) singularity as $x \rightarrow \xi$. Smoothing f increases convergence rate of the represented solution $u(x) = \int_\Omega G(x, \xi) f(\xi) d\xi$ by eigen-coefficient decay.

8 Transform methods for translation-invariant problems

On $\mathbb{R}^n$ (or tori with periodic boundary conditions), many operators are *convolution* operators, diagonalized by the Fourier transform.

Conventions. For $u \in \mathcal{S}(\mathbb{R}^n)$,

$$\widehat{u}(k) = \int_{\mathbb{R}^n} e^{-ik \cdot x} u(x) dx, \quad u(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{ik \cdot x} \widehat{u}(k) dk.$$

If L has constant coefficients, its *symbol* $L(k)$ satisfies $\widehat{Lu}(k) = L(k) \widehat{u}(k)$.

Green kernel via symbol. If $L(k)$ has no zeros (or after choosing a distributional prescription at zeros), the free-space Green's function is

$$\widehat{G}(k) = \frac{1}{L(k)}, \quad G(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \frac{e^{ik \cdot x}}{L(k)} dk,$$

interpreted as an oscillatory integral (tempered distribution).

Example 8.1 (Laplace). $L = -\Delta \Rightarrow L(k) = |k|^2$. Then

$$G(x) = \mathcal{F}^{-1}\left(\frac{1}{|k|^2}\right) = \begin{cases} \frac{1}{(n-2)\omega_n} |x|^{2-n}, & n \geq 3, \\ -\frac{1}{2\pi} \log |x|, & n = 2, \end{cases}$$

reproducing the fundamental solutions.

Example 8.2 (Helmholtz and radiation condition). $L = \Delta + k_0^2 \Rightarrow L(k) = -(|k|^2 - k_0^2)$. The inverse requires a causal/outgoing prescription:

$$\widehat{G}(k) = \frac{-1}{|k|^2 - k_0^2 - i0}.$$

Taking the inverse transform yields the outgoing kernels $G_3(x) = \frac{e^{ik_0|x|}}{4\pi|x|}$ and $G_2(x) = \frac{i}{4} H_0^{(1)}(k_0|x|)$.

Example 8.3 (Heat kernel (time domain)). For $u_t - \Delta u = \delta_{t=0} \delta_{x=0}$, the space-time Fourier transform gives $\widehat{G}(k, \omega) = \frac{1}{-i\omega + |k|^2}$, hence

$$G(t, x) = \frac{\mathbf{1}_{t>0}}{(4\pi t)^{n/2}} e^{-\frac{|x|^2}{4t}}.$$

Remark 8.4 (Periodic domains). On the torus $\mathbb{T}^n$, the spectrum is discrete: $k \in \mathbb{Z}^n$. Replace integrals by Fourier series,

$$G(x) = \sum_{k \in \mathbb{Z}^n \setminus \{0\}} \frac{e^{ik \cdot x}}{L(k)},$$

with the $k = 0$ mode removed when necessary (e.g. Laplacian on $\mathbb{T}^n$).

Remark 8.5 (Regularization at zeros). If $L(k)$ vanishes (Laplace at $k = 0$, Helmholtz on the sphere $|k| = k_0$), define $1/L(k)$ as a tempered distribution by limiting absorption ($\pm i0$), principal value, or by projecting off the nullspace; the physical choice is fixed by boundary/radiation conditions.

Summary. For translation-invariant operators, the transform route yields G in one line via $1/L(k)$. For bounded/curved domains, combine free-space G with images, boundary integral equations, or eigenfunction expansions from [Section 7](#).

9 How to select a construction (practical playbook)

- (1) **1D with separated BCs:** use the Sturm–Liouville recipe; two homogeneous bases + Wronskian.
- (2) **Simple *flat* boundaries (planes):** start from free-space kernel and enforce BCs by images.
- (3) **2D *smooth* simply-connected domains:** conformally map to the disk or half-plane and pull back the disk kernel.
- (4) **Helmholtz/Time-harmonic:** pick the *outgoing* free-space Φ (Sommerfeld radiation condition); use images only when symmetry/planarity allows.

10 Worked micro-examples

Example 10.1 (Poisson on $(0, 1)$ with $f \equiv 1$). Using the Dirichlet G from §2,

$$u(x) = \int_0^1 G(x, \xi) d\xi = \frac{1}{2}x(1 - x).$$

Example 10.2 (Half-space electrostatics). For a unit point charge at ξ above a grounded plane $x_3 = 0$, the potential is $u(x) = G(x, \xi)$ from §4. Field lines and image charge at ξ^* satisfy Gauss law and zero trace.

11 Numerical notes (for viewers and plotting)

- In 2D, treat the logarithmic singularity by subtracting/adding $-\frac{1}{2\pi} \log |x - \xi|$ and integrating the smooth remainder; or integrate in polar coordinates around ξ .
- For Helmholtz in 2D, avoid evaluating $H_0^{(1)}$ too near 0; use series expansions or asymptotics depending on kr .
- In 1D plots, enforce the continuity of G and the unit jump in $p \partial_x G$ at $x = \xi$.

A Quick kernel table

Operator / Domain	Green / Fundamental solution
$-\Delta$ in $\mathbb{R}^3$	$\frac{1}{4\pi r}$
$-\Delta$ in $\mathbb{R}^2$	$-\frac{1}{2\pi} \log r$
$(\Delta + k^2)$ in $\mathbb{R}^3$	$\frac{e^{ikr}}{4\pi r}$ (outgoing)
$(\Delta + k^2)$ in $\mathbb{R}^2$	$\frac{i}{4} H_0^{(1)}(kr)$ (outgoing)
Dirichlet on $\mathbb{D}$	$-\frac{1}{2\pi} \log \left \frac{z - \xi}{1 - \bar{\xi}z} \right $
Dirichlet on $\mathbb{H}$	$-\frac{1}{2\pi} \log \left \frac{z - \xi}{z - \bar{\xi}} \right $
Dirichlet on $(0, 1)$, $-y''$	$\min(x, \xi)(1 - \max(x, \xi))$

B Minimal TikZ stubs (optional figures)

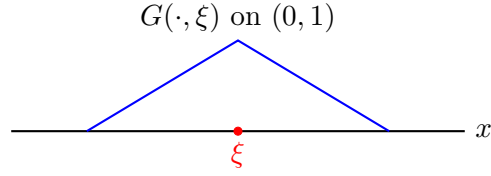


Figure 1: Schematic shape of a 1D Dirichlet Green function.